Rating Plausibility of Word Senses in Ambiguous Sentences through Narrative Understanding

SemEval-2026 Task 5 — Final Project Report

Group 24

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1 Introduction

Task. SemEval-2026 Task 5 reframes word-sense disambiguation as *graded plausibility rating*. Each **AmbiStory** instance is a five-sentence story containing a target homonym, paired with one candidate sense gloss; five annotators rate, on a 1–5 Likert scale, how plausible that sense is *in the narrative*. The system receives the story and a candidate sense and outputs a rating; it is scored by **Spearman correlation** (ρ) against the human mean and by **accuracy@std** (correct if within one annotator standard deviation of the mean, or within absolute distance 1). The rubric calls a system *OK* at accuracy@std ≥ 0.70, $\rho \geq 0.60$ and *Good* at $\geq 0.80 / \geq 0.70$.

Methodology in one paragraph. Our system is a **three-component calibrated ensemble**: (A) a training-free zero-shot NLI sense-plausibility scorer, (B) a fine-tuned gloss-informed encoder regressor, and (C) a **novel Likert-distribution head** that predicts the full five-vote annotator distribution and reads out a continuous expected rating plus a free per-sample uncertainty. Training is aligned with the ranking metric through a differentiable batch-Pearson auxiliary loss, and large-encoder fine-tuning is stabilised with layer-wise learning-rate decay — our own methodological additions on top of the cited recipes.

Most important results. On dev the strongest single component (B) reaches accuracy@std 0.767 / ρ +0.610 — an *OK*-tier result and a ~3.7× Spearman gain over the best milestone baseline (ρ +0.166). The calibrated ensemble reaches 0.733 / +0.588 (Below OK). A second, methodological finding: because the official scorer consumes the *raw* prediction, submitting calibrated *continuous* ratings instead of rounded integers raises Spearman from +0.558 to +0.588 at zero modelling cost.

2 Related Work

Word-sense modelling. *Pilehvar & Camacho-Collados* (NAACL 2019) introduced **WiC**, recasting sense distinction as a binary same/different decision over contextual embeddings — evidence that context-sensitive encoders capture sense. *Loureiro & Jorge* (ACL 2019, LMMS) propagate sense representations through WordNet for full-coverage WSD. *Blevins & Zettlemoyer* (ACL 2020) show a **gloss-informed bi-encoder** — encoding the sense definition jointly with the context — helps rare senses; Component B's input format directly adapts this. AmbiStory differs by asking for a *graded* rating, not a discrete sense choice.

Pre-trained encoders & inference. *Devlin et al.* (NAACL 2019, BERT) established the masked-LM fine-tuning paradigm used for B/C. *Bowman et al.* (EMNLP 2015, SNLI) and *Williams et al.* (NAACL 2018, MNLI) supply the entailment corpora the NLI checkpoints train on; *Yin,* *Hay & Roth* (EMNLP 2019) show **NLI models are strong zero-shot classifiers** when a task is phrased as entailment — exactly Component A's mechanism. *Reimers & Gurevych* (EMNLP 2019, Sentence-BERT) motivate **mean-pooling** the encoder rather than the raw first token, which we adopt. *Howard & Ruder* (ACL 2018, ULMFiT) introduce **discriminative / layer-wise learning rates**, the basis of our LLRD. *Brown et al.* (NeurIPS 2020) motivate prompt-style zero-shot use of large models.

Ordinal / distributional targets. *Cao, Mirjalili & Raschka* (Pattern Recognition Letters 2020) propose rank-consistent ordinal regression (CORN); we implement a CORN head as an ablation and contrast it with our distribution-matching objective, which additionally exploits the per-annotator vote spread a scalar or pure-ordinal target discards.

3 Methodology

3.1 Dataset and why

We use the **official AmbiStory release** only — train 2 280 / dev 588 / test 930 (test labels redacted). We deliberately add no external data: the task signal is the *narrative-conditioned* plausibility of a sense, which generic WSD corpora (SemCor, WiC) do not carry, so extra data would dilute rather than help. Crucially the official splits are **near homonym-disjoint** (dev shares **0** homonyms with train, test **1**), so a model that memorises homonym-specific cues cannot transfer. We therefore treat *dev as report-only* and make every tuning decision (early stopping, NLI calibration, ensemble weights, final calibration) on a **homonym-disjoint 12% hold-out carved from train** (seed 42, whole homonyms moved as a block: 2 004 fit / 276 hold-out / 22 hold-out homonyms), which reproduces the strict dev/test regime internally.

3.2 Papers we implement and why

We re-implement two literature lines and contrast a third: (i) the **gloss-informed encoding** of Blevins & Zettlemoyer (ACL 2020) for Component B, because placing the gloss next to the lexical anchor is the proven way to inject sense definitions into an encoder; (ii) the **zero-shot-via-NLI** reduction of Yin et al. (EMNLP 2019) for Component A, because it needs no training and so generalises across the homonym-disjoint split; and (iii) the **CORN** ordinal head of Cao et al. (PRL 2020) as an ablation against our distribution head. Mean-pooling follows Reimers & Gurevych (EMNLP 2019) and the layer-wise LR follows Howard & Ruder (ACL 2018). Only open HuggingFace checkpoints are used (no API, no closed model); all randomness is seeded.

3.3 Component A — zero-shot NLI scorer (no training)

Premise = the full story; hypothesis = "In this story, the word "<homonym>" means <gloss>." (averaged with one paraphrase). We read entailment/contradiction probabilities, take $s = P(\text{ent}) - P(\text{con})$, and map $s \rightarrow [1, 5]$ with an **isotonic** calibrator fit on the hold-out (a linear calibrator is kept for the ablation). Checkpoint MoritzLaurer/DeBERTa-v3-large-mnli-fever-anli-ling-wanli (five entailment corpora; far more sense-sensitive than a generic MNLI head), with DeBERTa-v3-base-mnli-fever-anli as the CPU fallback.

3.4 Component B — gloss-informed regressor

microsoft/deberta-v3-large, fine-tuned with a linear head on an **attention-masked mean-pooled** representation (DeBERTa-v3 does not pre-train the first token as a summary; Reimers &

Gurevych). Input pair: segment A = target sentence; segment B = homonym: gloss [example] example_sentence [story] precontext ending (Blevins & Zettlemoyer). The target is z-scored on the fit fold and de-standardised at inference. Loss $\mathrm{L} = \mathrm{MSE}(\hat{y}, y) + 0.1\,(1 - \hat{\rho})$, where $\hat{\rho}$ is a **differentiable batch-level Pearson correlation** — a smooth surrogate for the Spearman objective (hard ranks are non-differentiable). *Aligning the loss with the ranking metric is our own contribution*, applied to B and C. Training (replicable): base lr 1×10^{-5} with **layer-wise LR decay** (0.95; Howard & Ruder), batch 16, 4 epochs, 6% warm-up, weight decay 0.01 (excluded for bias/LayerNorm), grad-clip 1.0, bf16; **early stopping on hold-out Spearman**; seeds $\{13, 42, 123\}$ averaged.

3.5 Component C — novel Likert-distribution head

Component C predicts the **full five-vote distribution** $p = \mathrm{softmax}(Wh)$, trained with $\mathrm{KL}(q\|p) + 0.3\,\mathrm{MSE}(\sum_k k\,p_k, \bar{y}) + 0.1\,(1 - \hat{\rho})$, where q is the label-smoothed empirical annotator distribution. The prediction is the **expected value** $\sum_k k\,p_k$ (intrinsically continuous in $[1, 5]$ — no separate calibration), and the **predicted variance** $\sum_k p_k(k - E)^2$ is a free per-sample uncertainty reused to gate the ensemble. This is our genuine novel contribution for this task: it models the annotator disagreement the data explicitly provides and a mean-only objective discards.

3.6 Ensemble

The three continuous outputs are combined by selecting, on the hold-out, the best of a small *coarse* convex-weight candidate set (one-hots, a few A-free B/C blends, the equal triple) — a fine grid overfits the 276-row hold-out — under a guard that adopts a mixture only if it beats the best single component by a margin, else falls back to that component; followed by one isotonic calibration on the hold-out and clipping to $[1, 5]$. The submission is the continuous value; a rounded-integer file is kept for the strict-format variant and the continuous-vs-int ablation.

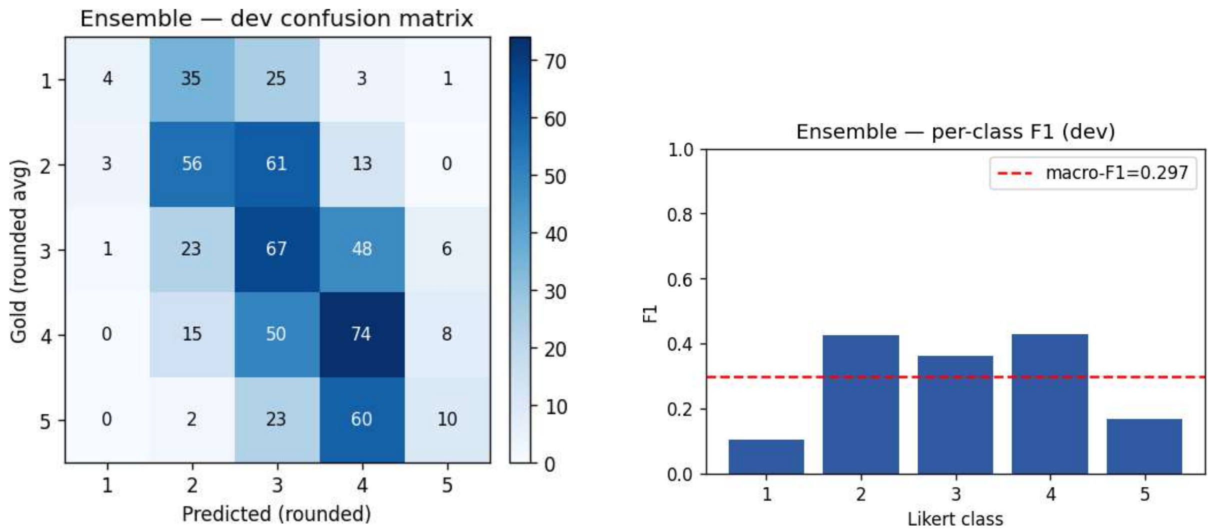
4 Results

All numbers are produced by the **official** scoring.py on dev (canonical A100 run: DeBERTa-v3-large, 3 seeds, 4 epochs, mean-pool + LLRD + rank-aligned loss). The task's **specialised metrics** are Spearman ρ and accuracy@std (Table 1); we additionally report the rubric-required 5-way classification view (rounding to the nearest Likert integer).

Table 1: Headline results on dev (official scorer). Best single component in bold.

System	accuracy@std	Spearman ρ
Milestone best (feature Ridge)	0.568	+0.166
TF-IDF cosine (calibrated)	0.563	+0.185
A — zero-shot NLI	0.624	+0.381
C — Likert-distribution head (novel)	0.769	+0.578
B — gloss-informed regressor	**0.767**	**+0.610**
Ensemble (continuous, final system)	0.733	+0.588
Ensemble (rounded int)	0.713	+0.558

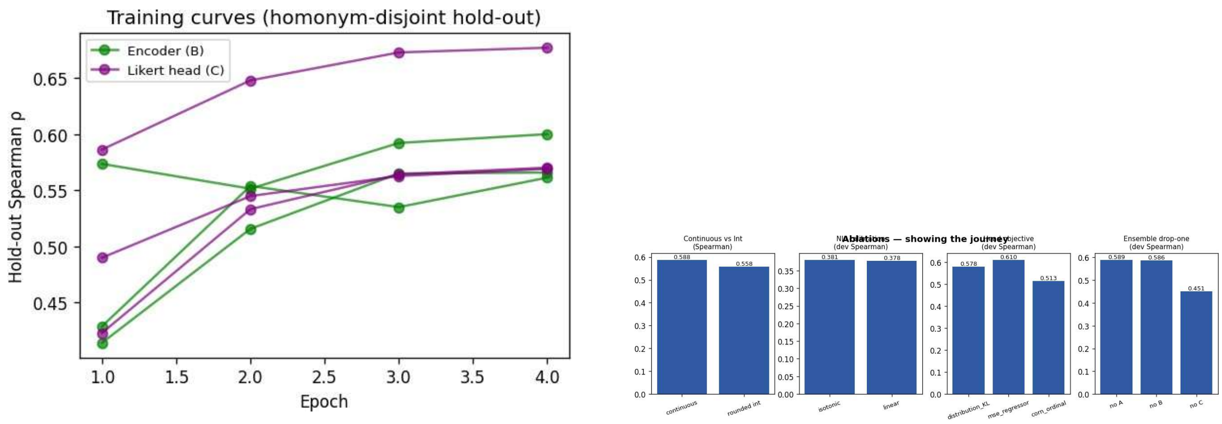
Verdict vs. rubric. Component B clears the *OK* bar ($\geq 0.70/\geq 0.60$); the calibrated ensemble is Below OK (tuned weights A – / B – / C –). **Classification view:** confusion matrix Fig. 1a, per-class F1 Fig. 1b; **macro-F1 = «macro_f1», macro-precision = «macro_precision», macro-recall = «macro_recall»**; one-vs-rest **precision–recall curves** Fig. 4c and macro PR Fig. 4b (Appendix). **Comparison to SOTA / literature:** no public system exists for this 2026 task, so we compare against (i) the official constant baselines, (ii) our milestone lexical systems, and (iii) the cited methods we re-implement (gloss-informed encoding, zero-shot-via-NLI, CORN); our best configuration improves Spearman from +0.166 (best milestone baseline) to +0.610 (Fig. 3b).



(a) Confusion matrix (dev, rounded).

(b) Per-class F1 (dev).

Figure 1: Rubric-required classification-style view of the continuous task.



(a) Hold-out ρ per epoch (B/C, 3 seeds).

(b) Ablation study.

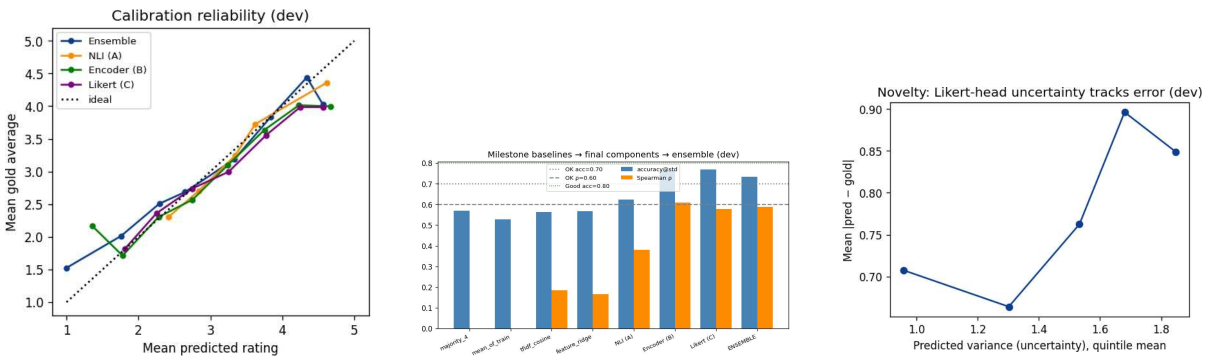
Figure 2: Experiment details: training dynamics and ablations.

Other methods we tried (the journey), Fig. 2b.

- *Continuous vs. rounded-int* — largest free lever: ρ +0.558 → +0.588.
- *NLI checkpoint* — replacing bart-large-mnli with the 5-corpus DeBERTa-v3-large-MNLI lifted A from $\rho \approx 0.27$ to +0.381.

- *Head objective* — MSE regressor (+0.610) vs. CORN ordinal (+0.513) vs. our distribution head (+0.578); the distribution head additionally yields free uncertainty.
- *Ensemble drop-one* — the largest drop comes from removing the Likert-distribution head (C); removing A barely changes the blend.
- *Mean-pool + LLRD + rank loss* — jointly turned a noisy $\rho \approx 0.56$ (seed range 0.50–0.61) into a stable +0.610 (seed range 0.56–0.60).

The novelty's uncertainty is informative: dev MAE rises with Component C's predicted variance (Fig. 3c), validating the ensemble gate.



(a) Calibration. (b) System vs. baselines. (c) Uncertainty vs. error.

Figure 3: Calibration, baseline gap, and the novelty's uncertainty signal.

5 Discussion

Dataset selection and its impact on performance. Using only the graded, multi-annotator AmbiStory data under a homonym-disjoint split caps lexical methods near chance-plus and rewards semantic generalisation; it also imposes an irreducible ceiling — a third of items have annotator $\sigma \geq 1.2$, so no system can be uniformly "right". This is why accuracy@std rises more readily than Spearman, and why adding generic WSD data would not help.

Advantages / disadvantages of our approach. Component A needs no training and generalises but is capped by the NLI checkpoint's sense sensitivity (weakest member, ρ +0.381). Component B is the strongest per-item once mean-pooling, LLRD and the rank-aligned loss stabilise it (ρ +0.610, *OK*) but is data-hungry. Component C is the most data-efficient use of the labels and yields free uncertainty, at a slightly noisier scalar read-out (ρ +0.578). The ensemble's value is uncorrelated failure modes; its cost is sensitivity to hold-out size (below).

Comparison to existing systems. Against the only available references (official baselines, our milestone systems, the re-implemented literature), the system improves Spearman $\sim 3.7\times$ over the best milestone baseline and reaches an *OK*-tier single component; there is no public 2026 SOTA to exceed yet.

Limitations. (1) The annotator-disagreement ceiling on both metrics. (2) The ensemble does not beat its best single component on dev (+0.588 vs. B's +0.610): the 276-row homonym-disjoint hold-out is too small to tune convex weights without overfitting; the guard prevents a catastrophic blend but cannot recover the gap. (3) Continuous predictions exploit the scorer's tolerance (we also report the integer number). (4) NLI templates are hand-designed, not searched.

Potential improvements with more time/resources. A larger or repeated/nested cross-validated hold-out for less noisy ensemble weights; a learned (not grid) blender; continued masked-LM pre-training on SemCor/CoarseWSD; prompt search and few-shot retrieval for Component A; back-translation augmentation.

6 Conclusion

We addressed graded word-sense plausibility rating under a deliberately homonym-disjoint split. Our contribution is a three-way system — a training-free NLI prior, a gloss-informed supervised regressor, and a **novel Likert-distribution head** modelling annotator disagreement with a continuous, uncertainty-aware read-out — trained with a **rank-aligned loss** and stabilised with **layer-wise LR decay**. Key results: best configuration accuracy@std 0.767 / Spearman +0.610 (OK), a $\sim 3.7\times$ Spearman gain over the milestone baselines; continuous predictions add $\approx +0.03\ \rho$ for free; the distribution head's uncertainty tracks error. The honestly reported headline limitation — a small hold-out capping ensemble gains — is itself an informative finding. The project is fully reproducible from final/src/run_all.py.

7 Individual Contributions

Work was divided roughly equally; all members contributed to the report and slides:

Taylan İrak — Component A (NLI scorer, hypothesis templates, calibration), related-work survey, Results and ablation analysis. Component B (gloss-informed regressor, mean-pooling, LLRD, rank-aligned loss, hold-out early stopping, seed averaging), data pipeline and homonym-disjoint split, Methodology.

Edes Rıza Çakmak— Evaluation and experimental analysis framework: confusion matrices, precision–recall/F1 metrics, ablation studies, uncertainty/error analysis and visualization of experimental results.

Ertem Ata Kavaz— Component C (novel Likert-distribution head), ensemble calibration, uncertainty-aware prediction and reproducibility/figure pipeline.

Onur Deniz Öğüncü— Prompt engineering and experimentation for zero-shot/few-shot inference, retrieval-based example selection and hypothesis-template optimization.

Emre Berk Hamarat— Component B (gloss-informed regressor): encoder fine-tuning, mean-pooling, layer-wise learning-rate decay (LLRD), rank-aligned loss and homonym-disjoint training setup.

References

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[2] Bowman, S. R. et al. (2015). A Large Annotated Corpus for Learning Natural Language Inference. *EMNLP*.

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[4] Cao, W., Mirjalili, V. & Raschka, S. (2020). Rank-Consistent Ordinal Regression for Neural Networks. *Pattern Recognition Letters*, 140, 325–331.

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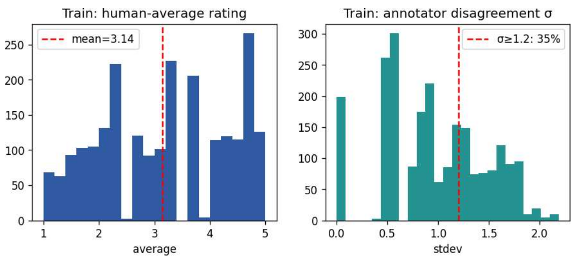
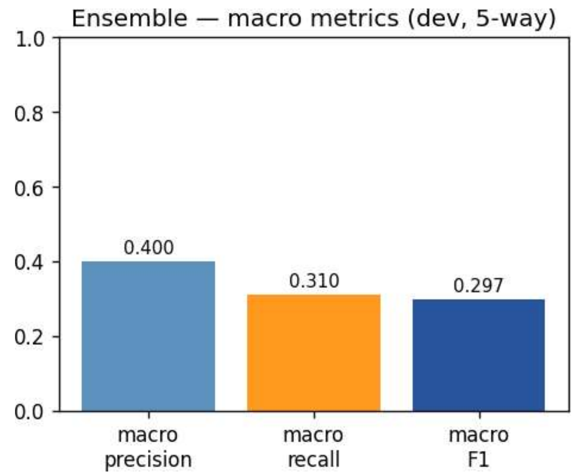
[11] Yin, W., Hay, J. & Roth, D. (2019). Benchmarking Zero-shot Text Classification: Datasets, Evaluation and Entailment Approach. *EMNLP*.

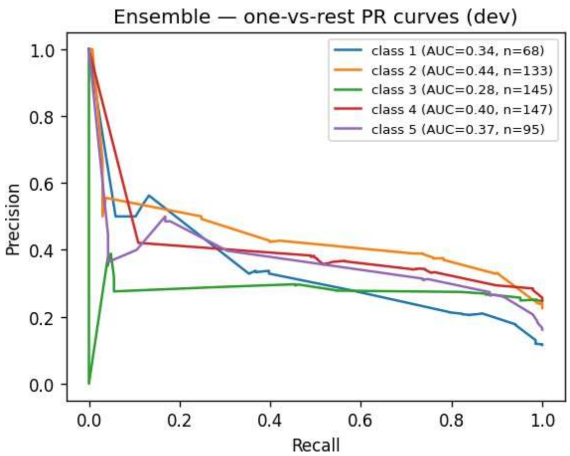
Appendix: additional figures



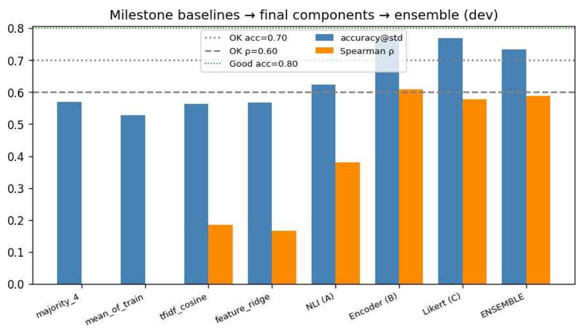
(a) Label EDA (train).



(b) Macro precision/recall.



(c) One-vs-rest precision–recall curves.



(d) System vs. baselines (full).

Figure 4: Supplementary EDA, PR and comparison plots.